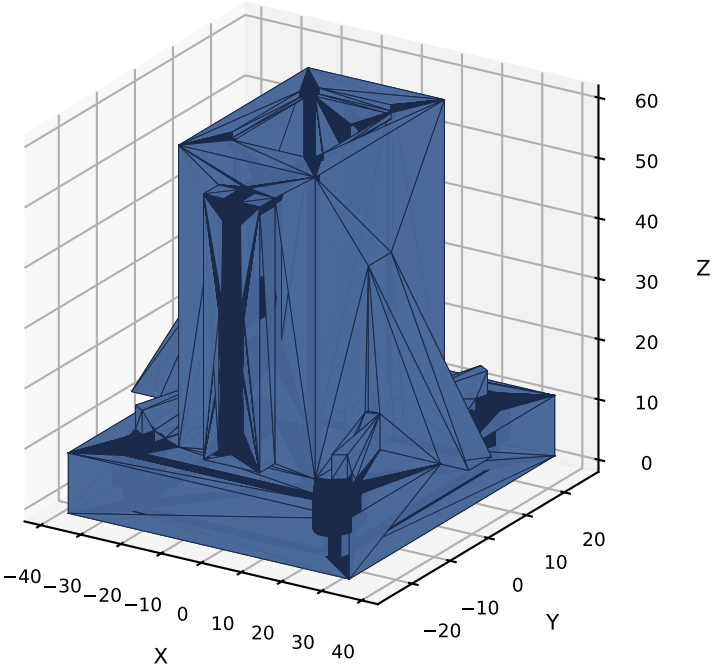


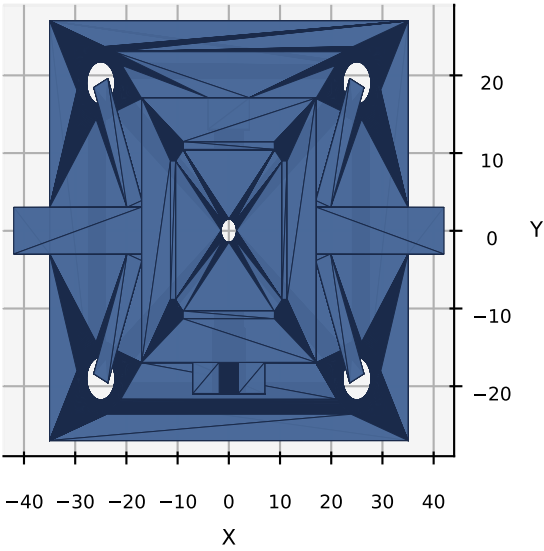
# CSM V3 -- HMI Module 10 -- Mast Base Socket V1.3

FREEZE candidate: lead-in chamfer, drain hole, 4 diagonal ribs, FRONT boss for inserts, wall-check verified

Isometric View

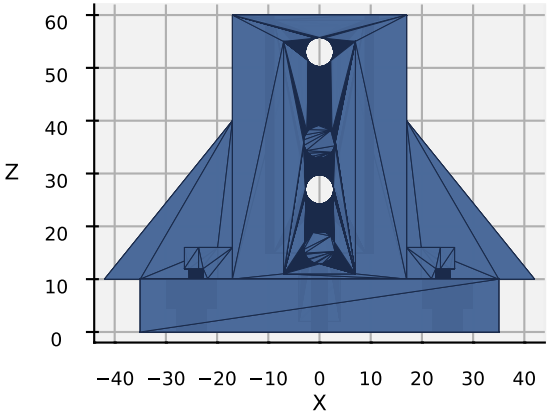


Top View (X-Y) -- ribs + bolt counterbores

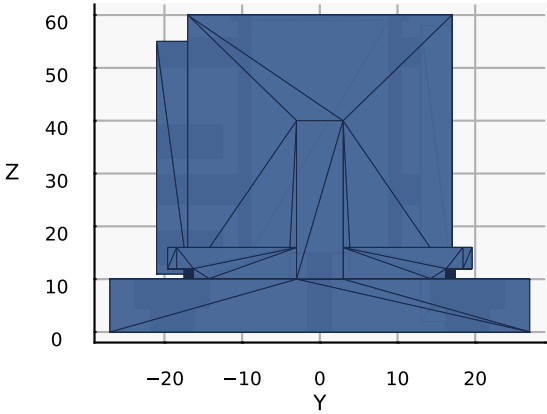


|                        |                       |
|------------------------|-----------------------|
| <b>PART</b>            | Mast Base Socket      |
| <b>VERSION</b>         | V1.3                  |
| <b>MODULE</b>          | 10 -- HMI             |
| <b>QTY</b>             | 2 (left + right)      |
| <b>MATERIAL</b>        | PETG (PA12 prod)      |
| <b>INFILL</b>          | 100%, 4 walls         |
| <b>LAYER</b>           | 0.2 mm                |
| <b>MASS</b>            | ~101 g each           |
| <b>TIME</b>            | ~4 h each             |
| <b>INSTALL</b>         | (X=+/-75, Y=-210)     |
| <b>DIMENSIONS (mm)</b> |                       |
| Socket outer           | 34.0 sq x 50.0        |
| Pocket                 | 20.8 sq, R1.5 fillets |
| Wall thick             | 6.6                   |
| Foot                   | 70.0 x 54.0 x 10.0    |
| Foot bolt CB           | 9.5 x 5.5             |
| Gusset                 | 25.0 x 30.0 x 6       |
| Anti-creep rib         | 2x 3.0 x 6.0          |
| Cable channel          | 8.0 x 4.0 rear        |
| Insert pocket          | 6.2 x 8.5             |
| Insert pitch           | 4 @ 20.0              |

Front View (-Y) -- 4x insert pockets @ 20 mm



Side View (+X) -- larger gusset + rib + cable channel



## V1.3 IMPROVEMENTS

1. R1.5 pocket inner fillets (printable + easier 2020 insertion + lower stress)
2. Gussets enlarged 18x22 -> 25x30
3. Foot thickness 8 -> 10 mm
4. 2x anti-creep ribs (3 x 6) on foot top
5. Foot bolt CSK -> counterbore (socket-head)
6. Insert pitch 10 -> 20 mm (std patterns)
7. cut\_cylinder() helper in macro

### ASSEMBLY (per bracket):

1. Print PETG foot-down, 100% infill, 4 walls
2. Heat 4x M5 brass inserts into FRONT pockets
3. 2x M5 T-nuts into 2020 FRONT slot
4. Slide bracket pocket down over 2020 base
5. M5 x 12 socket-heads through FRONT into T-nuts
6. Bracket on wood at (X=+/-75, Y=-210)
7. Drill 4x dia 5.5 through wood under bolt grid
8. M5 x 30 socket-heads through foot CB into wood
9. Washer + M5 nyloc nut under wood; tighten
10. Verify 2020 stands square in both planes